

Knee MRI — 30 Sample Cases

A worked teaching file of the 30 most common knee MRI diagnoses with realistic dictated findings and impressions.

Meniscus

Case 1 — Medial meniscus posterior horn horizontal/oblique tear

Clinical: 54-year-old man, gradual medial joint-line pain after squatting at work, no acute injury.

F: On sagittal and coronal PD fat-sat images there is a horizontal-oblique high-signal line within the **posterior horn of the medial meniscus** reaching the inferior articular surface, measuring approximately 11 mm in AP extent. A small associated **parameniscal cyst** is seen tracking medially measuring 8 x 5 mm. No displaced fragment. Mild adjacent grade 2 chondral signal change over the medial femoral condyle without full-thickness loss. The ACL, PCL, and collateral ligaments are intact. No significant joint effusion.

I: Degenerative horizontal-oblique tear of the medial meniscus posterior horn with small parameniscal cyst.

Case 2 — Radial tear of the medial meniscus

Clinical: 47-year-old woman, twisting injury stepping off a curb, mechanical medial pain.

F: A vertically oriented full-thickness cleft perpendicular to the long axis of the **medial meniscus** is identified at the junction of the body and posterior horn, producing the **truncated-triangle** and **ghost meniscus** signs on consecutive sagittal images. The radial defect spans the full 9 mm meniscal width. Early 2 mm meniscal extrusion at the medial margin. No bucket-handle displacement. Cruciate and collateral ligaments intact. Trace effusion.

I: Full-thickness radial tear of the medial meniscus body–posterior horn junction.

Case 3 — Medial meniscus posterior root tear with extrusion

Clinical: 61-year-old woman, felt a pop rising from a chair, acute medial pain.

F: There is a radial-type defect at the **posterior root attachment of the medial meniscus** with a fluid-filled cleft and the **ghost meniscus sign** at the root on sagittal images. The meniscal body is extruded 5 mm beyond the medial tibial margin on the coronal sequence. Associated **subchondral marrow edema** within the medial femoral condyle and medial tibial plateau without discrete fracture line. Mild medial compartment cartilage thinning. ACL intact.

I: Medial meniscus posterior root tear with significant meniscal extrusion, functionally equivalent to a complete tear and predisposing to accelerated medial compartment degeneration.

Case 4 — Bucket-handle tear with displaced fragment and double-PCL sign

Clinical: 23-year-old male soccer player, locked knee unable to fully extend after a tackle.

F: The **medial meniscus** is foreshortened with absence of the normal bow-tie configuration on peripheral sagittal images. A displaced meniscal fragment lies within the intercondylar notch anterior and inferior to the PCL, producing the **double-PCL sign** on the sagittal sequence and the **fragment-in-notch** appearance on coronal images. The donor posterior horn is diminutive. Moderate joint effusion. ACL intact.

I: Displaced bucket-handle tear of the medial meniscus with intercondylar fragment (double-PCL sign), correlating with the locked knee.

Case 5 — Lateral meniscus tear

Clinical: 29-year-old man, pivoting basketball injury, lateral joint-line tenderness.

F: Vertical-longitudinal increased signal reaching the inferior surface of the **posterior horn of the lateral meniscus** measuring 7 mm, situated just lateral to the popliteus tendon hiatus (a normal structure, not to be mistaken for tear). No displaced fragment and no double-PCL sign. The lateral meniscus body and anterior horn are normal. Small reactive lateral compartment effusion. ACL intact.

I: Vertical-longitudinal tear of the lateral meniscus posterior horn.

Case 6 — Discoid lateral meniscus with tear

Clinical: 16-year-old girl, recurrent lateral clicking and intermittent pain, no major trauma.

F: The **lateral meniscus** demonstrates a continuous bow-tie configuration across more than three contiguous 4 mm sagittal slices and a widened transverse diameter exceeding 15 mm on coronal images, consistent with a **discoid lateral meniscus** (complete type). Within the thickened meniscus there is horizontal intrasubstance high signal reaching the inferior surface indicating a degenerative tear. No displacement. Cruciates intact.

I: Complete discoid lateral meniscus with a horizontal tear.

Case 7 — Degenerative meniscal maceration

Clinical: 68-year-old man, chronic medial pain, prior known osteoarthritis.

F: The **medial meniscus** posterior horn and body are diffusely irregular, blunted, and macerated with complex high signal and loss of the normal triangular morphology; no discrete reparable tear plane. Meniscal extrusion measures 4 mm medially. Background medial compartment full-thickness cartilage loss and subchondral cystic change. Marginal osteophytes. Cruciates intact.

I: Degenerative maceration of the medial meniscus in the setting of established medial compartment osteoarthritis.

Cruciate Ligaments

Case 8 — Acute complete ACL tear with pivot-shift contusions and Second fracture

Clinical: 19-year-old female soccer player, non-contact pivot with a pop and immediate swelling.

F: The **anterior cruciate ligament** is discontinuous with absent fibers in the mid-substance, a wavy redundant distal stump, and increased T2 signal; the expected oblique orientation is lost. Characteristic **pivot-shift bone contusions** involve the lateral femoral condyle (sulcus terminalis) and posterolateral tibial plateau. A small cortical avulsion at the lateral tibial margin represents a **Segond fracture**, indicating anterolateral ligamentous injury. Anterior tibial translation and uncovering of the posterior horn lateral meniscus are noted. Large hemarthrosis.

I: Acute complete ACL tear with classic pivot-shift contusion pattern and Segond fracture.

Case 9 — ACL mucoid degeneration

Clinical: 58-year-old woman, posterior knee fullness and pain at terminal flexion, no instability.

F: The **anterior cruciate ligament** is diffusely thickened with ill-defined intermediate T2 signal interspersed between intact, taut fibers producing a **celery-stalk appearance**; the fibers remain continuous from origin to insertion. Adjacent intraosseous **ganglion/mucoid cysts** are present at the tibial insertion. No pivot-shift contusions and no anterior tibial translation. Menisci intact.

I: Mucoid degeneration of the ACL; no discrete tear or instability findings.

Case 10 — ACL reconstruction graft evaluation

Clinical: 31-year-old man, 2 years post ACL reconstruction, recurrent giving-way.

F: A **bone–patellar tendon–bone ACL graft** courses through the femoral and tibial tunnels. The intra-articular graft fibers are partially attenuated with intermediate signal and mild laxity; there is interval widening of the tibial tunnel and a focal full-thickness fiber discontinuity proximally suggesting **partial graft tear**. No cyclops lesion impinging on extension. Tunnel positions acceptable. Menisci intact.

I: Partial tear/attenuation of the ACL graft with tibial tunnel widening; clinical correlation for recurrent instability.

Case 11 — PCL tear

Clinical: 36-year-old man, dashboard injury in a motor-vehicle collision, posterior knee pain.

F: The **posterior cruciate ligament** is markedly thickened and edematous with intrasubstance high T2 signal and partial fiber disruption at the mid-substance; AP diameter measures 9 mm (upper limit normal ~7 mm). The femoral and tibial attachments remain intact, favoring an interstitial/partial tear rather than complete avulsion. Associated anterior tibial bone contusion. ACL intact. No posterior tibial avulsion fragment.

I: Acute high-grade partial (interstitial) PCL tear without bony avulsion.

Case 12 — O'Donoghue triad (ACL + MCL + medial meniscus)

Clinical: 25-year-old male skier, valgus + external rotation injury, immediate effusion.

F: Complete mid-substance tear of the **anterior cruciate ligament** with pivot-shift contusions of the lateral femoral condyle and posterolateral tibial plateau. The **medial collateral ligament** shows a high-grade sprain with periligamentous edema and partial proximal fiber disruption (grade 2–3). A peripheral vertical-longitudinal tear of the **medial meniscus** posterior horn is present at the meniscocapsular junction. Large hemarthrosis.

I: O'Donoghue unhappy triad: ACL tear, MCL injury, and medial meniscus tear.

Collaterals & PLC

Case 13 — MCL grade 2 sprain

Clinical: 21-year-old male football player, valgus blow to the lateral knee, medial tenderness.

F: The **medial collateral ligament** demonstrates periligamentous and intrasubstance edema with partial-thickness fiber disruption of the superficial component at its femoral origin; the deep fibers and overall ligament continuity are preserved. No complete discontinuity and no associated meniscal tear. Mild adjacent soft-tissue edema. ACL and PCL intact.

I: Grade 2 (partial) sprain of the superficial MCL at the femoral attachment.

Case 14 — LCL (fibular collateral ligament) tear

Clinical: 33-year-old man, varus injury with the foot planted, lateral instability.

F: The **fibular (lateral) collateral ligament** is thickened and discontinuous near its fibular head insertion with a lax, wavy distal stump and surrounding edema, consistent with a high-grade/complete tear. The **biceps femoris** insertion is also edematous. No fibular head avulsion fragment. Cruciates intact. Lateral meniscus intact.

I: High-grade tear of the fibular collateral ligament near the fibular insertion.

Case 15 — Posterolateral corner injury with arcuate sign

Clinical: 28-year-old man, hyperextension + varus injury, posterolateral pain and instability.

F: There is disruption of the **posterolateral corner** structures including the **fibular collateral ligament**, **popliteus tendon**, and **popliteofibular ligament** with diffuse periligamentous edema. A small avulsed cortical fragment at the **fibular styloid** represents the **arcuate sign**. Associated anteromedial femoral and tibial bone contusions reflect the hyperextension mechanism. Concurrent partial PCL injury is suspected.

I: Posterolateral corner injury with arcuate (fibular styloid avulsion) sign; assess for associated cruciate injury.

Extensor Mechanism & Patellofemoral

Case 16 — Patellar tendon partial tear

Clinical: 30-year-old recreational basketball player, anterior knee pain after jumping, palpable defect.

F: The **patellar tendon** is thickened at its proximal origin from the inferior patellar pole with intrasubstance high T2 signal and partial-thickness fiber disruption involving roughly 40% of the cross-section; the tendon remains in continuity. Adjacent **Hoffa fat-pad** edema and small inferior pole marrow edema. No complete rupture and no patella alta. No quadriceps tendon abnormality.

I: Proximal patellar tendon partial-thickness tear (jumper's knee), tendon in continuity.

Case 17 — Quadriceps tendon tear

Clinical: 63-year-old man, missed step on stairs, sudden inability to extend the knee.

F: The **quadriceps tendon** is completely disrupted approximately 1.5 cm above the superior patellar pole with a fluid-filled gap measuring 2.2 cm and retraction of the proximal tendon. There is **patella baja** from loss of the proximal tether and a moderate suprapatellar effusion with hemorrhage. The patellar tendon is intact. Mild background tendinosis at the torn margins.

I: Complete quadriceps tendon rupture with tendon gap and retraction; surgical repair indicated.

Case 18 — Acute lateral patellar dislocation with MPFL tear and bone-bruise pair

Clinical: 17-year-old girl, twisting non-contact injury during dance, sense of the kneecap "popping out."

F: Findings of transient **lateral patellar dislocation** with the characteristic **bone-contusion pair**: marrow edema involving the **inferomedial patellar facet** and the **anterolateral femoral condyle**. The **medial patellofemoral ligament (MPFL)** is disrupted at its patellar attachment with surrounding edema. A small osteochondral fragment (4 mm) is seen in the lateral gutter. Trochlear dysplasia and patella alta predispose. Large effusion/hemarthrosis.

I: Acute transient lateral patellar dislocation with MPFL tear, characteristic contusion pair, and a small osteochondral fragment.

Case 19 — Patellofemoral chondromalacia with maltracking

Clinical: 26-year-old woman, anterior knee pain with stairs and prolonged sitting, no trauma.

F: Axial PD fat-sat images show **chondromalacia** of the patellar median ridge and lateral facet with grade 3 fissuring and focal grade 4 full-thickness cartilage loss laterally. There is **lateral patellar tilt** (patellar tilt angle increased), a shallow trochlea, and an increased **TT–TG distance of 19 mm** indicating maltracking. Mild lateral retinacular thickening. No acute marrow edema.

I: Patellofemoral chondromalacia with lateral maltracking (increased TT–TG distance and patellar tilt).

Cartilage & Bone

Case 20 — Acute osteochondral injury

Clinical: 22-year-old man, acute impaction injury landing from a height, persistent pain.

F: A focal **osteochondral injury** of the lateral femoral condyle weight-bearing surface shows a 12 x 9 mm chondral fissure with underlying subchondral marrow edema and a partially detached osteochondral flap; fluid signal undermines the fragment indicating instability. A small intra-articular loose body is identified in the suprapatellar recess. Surrounding reactive effusion.

I: Acute unstable osteochondral injury of the lateral femoral condyle with a loose body.

Case 21 — Osteochondritis dissecans

Clinical: 15-year-old boy, chronic activity-related knee pain, occasional catching.

F: A well-corticated osteochondral lesion at the classic **lateral aspect of the medial femoral condyle** measures 15 x 12 mm. There is no high-signal fluid line fully encircling the fragment and the overlying cartilage is intact, indicating a **stable** lesion. No displaced fragment or loose body. Mild adjacent marrow edema. Physes open.

I: Stable osteochondritis dissecans of the medial femoral condyle (juvenile form).

Case 22 — SONK / subchondral insufficiency fracture

Clinical: 67-year-old woman, sudden spontaneous medial knee pain, no trauma, osteopenic.

F: The **medial femoral condyle** weight-bearing surface shows a subchondral low-signal fracture line paralleling the articular surface with extensive surrounding **bone marrow edema** and a focal subchondral T2-hypointense area, consistent with a **subchondral insufficiency fracture (SONK pattern)**. Mild overlying chondral flattening without large defect. Adjacent medial meniscus posterior root tear with extrusion contributing to altered load. Joint effusion.

I: Subchondral insufficiency fracture of the medial femoral condyle (SONK), associated with medial meniscus root tear.

Case 23 — Tibial plateau contusion/fracture

Clinical: 44-year-old man, fall from a ladder onto a flexed knee, lateral pain and inability to bear weight.

F: The **lateral tibial plateau** demonstrates a depressed articular fracture with a vertically oriented split component and approximately 4 mm of articular depression (Schatzker II pattern); extensive surrounding marrow edema. A **lipohemarthrosis** with a fat-fluid level confirms intra-articular fracture. The lateral meniscus is displaced peripherally over the fracture margin. ACL intact.

I: Lateral tibial plateau split-depression fracture (Schatzker II) with lipohemarthrosis.

Case 24 — Tricompartmental osteoarthritis

Clinical: 71-year-old woman, chronic diffuse knee pain and stiffness, progressive over years.

F: There is **tricompartmental osteoarthritis** with full-thickness cartilage loss in the medial, lateral, and patellofemoral compartments, marginal **osteophytes**, subchondral sclerosis and cystic change, and degenerative maceration of both menisci with medial extrusion. Mild varus alignment. Moderate effusion with a **Baker cyst**. No acute fracture.

I: Advanced tricompartmental osteoarthritis with degenerative meniscal disease and a Baker cyst.

Other

Case 25 — Baker (popliteal) cyst

Clinical: 55-year-old woman, posterior knee fullness and occasional tightness.

F: A well-defined fluid signal collection arises between the **medial head of gastrocnemius** and the **semimembranosus** tendons, communicating with the joint, measuring 4.5 x 2.0 x 6.0 cm, consistent with a **Baker cyst**. Mild internal septation and a few loose bodies. No rupture or dissection into the calf and no surrounding edema to suggest leakage. Associated joint effusion and degenerative meniscal changes.

I: Uncomplicated Baker cyst arising at the gastrocnemius–semimembranosus bursa.

Case 26 — Pes anserine bursitis

Clinical: 59-year-old woman with type 2 diabetes, medial knee pain below the joint line.

F: A fluid collection deep to the conjoined **pes anserine** tendons (sartorius, gracilis, semitendinosus) along the anteromedial proximal tibia measures 3.2 x 1.1 cm with surrounding soft-tissue edema, consistent with **pes anserine bursitis**. No communication with the joint. The MCL is intact and uninvolved. Mild background medial compartment degenerative change.

I: Pes anserine bursitis.

Case 27 — Iliotibial band friction syndrome

Clinical: 34-year-old male long-distance runner, lateral knee pain at ~2 miles, worse downhill.

F: There is ill-defined edema in the soft tissues deep to the **iliotibial band** overlying the lateral femoral epicondyle, with the IT band itself mildly thickened. No discrete bursal fluid collection or intra-articular extension. The lateral meniscus, LCL, and popliteus are intact, excluding intra-articular causes of lateral pain.

I: Iliotibial band friction syndrome.

Case 28 — Chronic ACL-deficient knee

Clinical: 41-year-old man, remote untreated ACL injury, chronic instability and giving-way.

F: The **anterior cruciate ligament** is not identified in its expected course with only scarred residual fibers, and there is fixed **anterior tibial translation** with uncovering of the posterior horn of the lateral meniscus. Secondary degenerative changes include medial compartment cartilage loss and a chronic radial/degenerative tear of the medial meniscus. No acute marrow edema. Notch osteophytes are present.

I: Chronic ACL-deficient knee with secondary degenerative meniscal and chondral changes.

Case 29 — Medial meniscocapsular separation (ramp lesion)

Clinical: 20-year-old male footballer, ACL injury, persistent posteromedial pain.

F: At the posteromedial periphery there is fluid signal and disruption at the **meniscocapsular junction** of the medial meniscus posterior horn (**ramp lesion**) with adjacent posteromedial tibial marrow edema. This accompanies a complete **ACL tear** with typical pivot-shift contusions. The meniscal body itself is otherwise intact and non-displaced.

I: Medial meniscocapsular (ramp) lesion accompanying acute ACL tear; important to identify as it is easily missed and alters surgical management.

Case 30 — Hoffa fat-pad impingement syndrome

Clinical: 38-year-old woman, anterior knee pain with extension, no acute injury.

F: The superolateral aspect of the **infrapatellar (Hoffa) fat pad** demonstrates ill-defined edema-like high T2 signal and mild mass effect between the patellar tendon and lateral femoral condyle, consistent with **Hoffa fat-pad impingement**. The patellar tendon is intact without tear. No focal mass or hemorrhage. No effusion. Patellofemoral cartilage is preserved.

I: Superolateral Hoffa fat-pad impingement syndrome.